

Living Brain Explorer – Summer Project Blueprint

Project Goal

Build a biologically-inspired interactive brain simulation showing learning, memory, attention and decision making. Research-backed concepts only.

Final Outcome

Interactive visual simulation with neurons, pathways, activity signals, sliders for memory/attention/reward, experiments and observations.

Software Stack

Python, NumPy, NetworkX, Matplotlib, Pygame, Jupyter Notebook. Optional: Blender for 3D visuals.

Research Topics

Neuron basics, synapses, Hebbian learning, reward systems, memory, attention, reinforcement learning.

Weeks 1-2

Learn neuroscience basics and Python libraries.

Weeks 3-4

Build neurons and pathways with visual signal movement.

Weeks 5-6

Add learning, memory and attention systems.

Weeks 7-8

Build interactive interface and run experiments.

Biological Accuracy

Use simplified but evidence-based principles rather than random behavior.

CV Impact

Strong for research, cognitive science, AI and interdisciplinary profiles.